

Abstract	High capacity and low latency of optical connections are ideal for supporting current and future communication services, including 5G and beyond. Although some of those services are already secured at the packet layer using standard stream ciphers, like the Advanced Encryption Standard and ChaCha, secure transmission at the optical layer is still not implemented. To secure the optical layer, cryptographic methods need to be fast enough to support high-speed optical transmission and cannot introduce significant delay. Moreover, methods for key exchange, key generation, and key expansion are required, which can be implemented on standard coherent transponders. In this paper, we propose Light Path SECurity (LPsec), a secure cryptographic solution for optical connections that involves fast data encryption using stream ciphers and key exchange using Diffie–Hellman protocol through the optical channel. To support encryption of high-speed data streams, a fast, general-purpose pseudorandom number generator is used. Moreover, to make the scheme more secure against exhaustive search attacks, an additional substitution cipher is proposed. In contrast to the limited encryption speeds that standard stream ciphers can support, LPsec can support high-speed rates. Numerical simulation for 16 quadrature amplitude modulation (QAM), 32-QAM, and 64-QAM show that LPsec provides a sufficient security level while introducing only negligible delay.
Date	2022-04-01
Citation Key	IQBALLPsecFastSecureCryptographicSystemOptical2022
DOI	10.1364/JOCN.444398
URL	https://opg.optica.org/jocn/abstract.cfm?uri=jocn-14-4-278
Importance	Medium (10+ citations)
FWCI	0.97
Citation percentile	0.71
Cites	24
Cited by	14
Related to	10
Retracted	No
PubPeer Alerts	0
OpenReview Alerts	0
Peer Reviewed	Yes

12. Tor: The Second-Generation Onion Router

Author(s)	Roger Dingledine (Works count: 30, Citations count: 4372, H-index: 20), Nick Mathewson (Works count: 19, Citations count: 5438, H-index: 12), Paul Syverson (Works count: 186, Citations count: 13818, H-index: 48)
Item Type	Conference Paper
Date	2004-07-31
Citation Key	DINGLEDINETorSecondGenerationOnionRouter2004
DOI	10.21236/ADA465464
URL	https://www.usenix.org/conference/13th-usenix-security-symposium/tor-second-generation-onion-router
Importance	High (100+ citations)
FWCI	N/A
Citation percentile	N/A